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A Real-Time Approach for Detection and Correction of False Data in PMU Measurements

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ABSTRACT

Recent studies imply that despite having a high level of security features and accurate algorithms, Phasor Measurement Units (PMUs) are vulnerable to False Data Injection (FDI) in the measurements. These FDIs could be: 1) an intentional FDI Attack (FDIA) or 2) unintentional errors resulting from limited precision, communication medium, noise, etc. Traditional false data detection techniques that use SCADA measurements in conjunction with the PMU measurements are insufficient to detect malicious data in PMU measurements. Thus, a new approach is proposed in this paper, which can detect the presence of false data in the PMU measurements and correct them, without relying on the use of the SCADA measurements. An affine relationship between the sparsely placed PMUs, based on the AC model of the network, is statistically extracted by offline training of a support vector regressor. This relationship is further used for the real-time detection and correction of false data in the PMU measurements. The proposed approach is tested on the IEEE 118 bus system and the Northern Regional Power Grid (NRPG) of India for detecting intentional as well as unintentional false data in single and multiple PMU measurements. Numerical results, thus obtained after testing synthetic and real-world data, are also compared with existing approaches, which justify and prove the efficacy of the proposed method.

1. Introduction

The State Estimation (SE) exercise lets a system operator know about the system states (i.e., the voltage magnitudes and associated phase angles at all the buses) for the stable and secure operation of the power network [1]. The advent of synchrophasor technology has revolutionized the SE by making it extremely fast, robust, and highly accurate [2]. However, the high-cost associated with PMUs makes the replacement of all the SCADA meters with the PMUs infeasible. Hence, sparse placement of PMUs is being done so that full system observability is ensured while covering all the critical measurements, and the rest of the non-PMU buses are equipped with SCADA meters [3]. The data quality of these PMU measurements raises concerns; for example, typical PMU bad data ratio in California ISO ranges from 10% to 17% [4]. These PMU measurements generally get corrupted due to: i) intentional data corruption by a cyber attack or, ii) unintentional data corruption during digital data processing, storage, and retrieval stage [5–7].

Several researchers have investigated the false data issue in PMU measurements from the perspectives of cyber attack as well as signal processing errors, and have suggested different methods to handle this problem. In [8], a Bayesian-based approximated filter is proposed to detect FDI when the probability of information corruption is high. However, it is also reported that the computational burden and time complexity of the filter will increase in the case of frequent data-injection attacks, thus making it unsuitable for real-time application. Wavelet packet decomposition-based techniques [9] are also proposed for the detection of measurement corruption. Such time-window based methods are likely to raise false alarms for any corrupt measurement sample present in a given time window. In [10], the formulation of FDIA to introduce bad data in SE and its countermeasure using Principal Component Analysis (PCA) is proposed. However, the study is limited to consideration of SCADA measurements only. In [11], a data quality filter is proposed, which is a statistical tool and will not require SCADA data, state estimator values, or system topology information. This aids the data integrity and situational awareness tool of the Bonneville Power Administration (BPA) in dealing with the corruption in the PMU measurements. However, this filter for voltage phasor is based on voltage magnitudes only, i.e., it will not detect bad data in phase angles. Any voltage magnitude outside the range of [0.75, 1.25]*pu* is considered to have data quality issues. This heuristic range is based on the experiences of BPA. In [12], the use of machine learning is discussed for the detection of potential FDIA in the smart grid using classification techniques, where measurements are classified as

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attacked or secured. However, being a classification technique, Esmalifalak et al. [12] fails to locate the meter being attacked. It also fails to quantify the amount of false data in the associated measurement. Authors of [13] propose a sparse optimization-based technique for FDI detection, but only considering SCADA. Various cyber attack strategies to introduce false data in Remote Terminal Units (RTUs) and PMUs, as well as their impact on SE, are reviewed in [6]. A robust FDIA detector is also proposed in [14], which uses a limited number of secure PMUs and short-term measurement forecasting to detect FDIA. However, recent literature [5,6,14,15] also shows that the PMU measurements are not immune to cyber attacks. Authors of [14] propose the detection of false data in PMUs with the help of PCA. Higher-dimensional principal components are used to classify outliers caused by false data from those created by system disturbances. However, the applicability and behavior of the method proposed in [14] are not discussed for the scenario when multiple bad data points of the same PMU exist in one data window. A statistical consistency check is proposed in [16] to detect the imperfect false data injection using a set of PMUs. However, the assumptions that the PMUs are secured and will not be attacked, and the attacker will not be able to keep the measurement residue uncharged post-attack are not necessarily true in practice. Alternating Direction Method of Multipliers (ADMM) is used in [15] to recover the PMU measurements. The approach in [15] can be extended to correct the falsified states, once the FDIA is detected in PMU measurements. However, the selection of crucial parameters for ADMM is a challenge, and the choice of these parameters significantly affects the accuracy of the measurement recovery.

The literature review thus reveals that most of the false data detection techniques require SCADA measurements to detect false data in the PMU measurements. Also, recent literature shows that the PMU measurements are not fully secure against cyber attacks [17–19]. In the event of a FDIA on the PMU measurements, an attacker manipulates the measurements of PMU as well as the nearby SCADA meters (RTUs) in a way so that corrupt measurements pass the measurement residual based tests [20] at energy control center and the attack remains hidden [21,22]. This makes the traditional bad data detection, identification, and correction strategies, insufficient to detect intentional false data in PMU measurements. Similarly, the inherent high noise level in SCADA measurements, low sampling rate and inability of time-stamping makes the SCADA measurements unsuitable for bad data detection in sparsely placed PMU measurements, in the event of unintentional data corruption. Literature review also reveals that apart from cyber attacks, PMU measurements are also vulnerable to unintentional FDI due to faulty current or potential transformers, noise in the communication medium, GPS jamming, etc. [2,23,24]. The traditional measurement residue based detection techniques require redundancy in terms of the number of measurements. However, in the present scenario, this redundancy of measurements for PMUs is not guaranteed because of the sparse placement of PMUs as well as the significant difference in the sampling rates and refresh rates of RTUs and PMUs. Hence, this gives rises to the requirement of an alternative approach for the detection of intentional as well as unintentional false data in the PMU measurements, which neither relies on the SCADA measurements nor the network topology and parameters. Therefore, a Support Vector Regression (SVR) based real-time false data detection and correction technique, which only requires the PMU measurements, is proposed in this paper. The key contributions and features of the proposed approach are

- An approach for real-time detection of coordinated FDIA on PMU measurements and their subsequent correction is presented in this paper. This approach is based on the use of SVR and the AC model of the power network.
- The proposed approach can also detect and correct unintentional false data in PMU measurement, even when the measurements are noisy.
- The proposed approach does not require any information regarding SCADA measurements or network topology and works only by using the PMU measurements.
- An affine relationship between the voltage phasors of the sparsely placed PMUs is statistically established by the offline training of SVR (s), which otherwise is not possible through the use of the conventional deterministic methods without using the SCADA based measurements. Such an affine relationship is useful for the real-time implementation (testing) of the trained SVRs in the proposed approach for detection and correction of bad data in PMU measurements.

The rest of the paper is organized as follows. Section 2 briefly discusses the mathematical basis of SVR and FDI model in PMU measurements. The proposed methodology to detect and correct false data in PMU measurements is discussed in Section 3. The testing of the proposed approach on two test systems is discussed in Section 4. A comparison of the proposed approach with the related works in [8,9,11,15,16,23] is also given. Section 5 concludes the paper.

2. Preliminaries

2.1. Support Vector Regression (SVR)

SVR is a supervised machine learning method that exploits the concept of hyperplane for regression analysis. It is an extension of the Support Vector Machine (SVM), which was first developed for classification problems [25]. The key features of SVR [26] that make it better than other machine learning algorithms are as follows

1. Use of kernels functions that convert nonlinear problems into linear problems.
2. Absence of a local minima that guarantees a global optimal solution.
3. The sparseness of the solution that results in better learning and faster computations (after learning).
4. Capacity control, obtained by optimizing the tolerance.

These key benefits of SVR have motivated us to use SVR in the proposed methodology in Section 3. The basis of SVR is briefly discussed next.

SVR can estimate the functional dependence of a dependent variable y on a set of independent variables Θ. It assumes that a relationship between the independent and dependent variables can be given by a deterministic function f with some additive noise as

$$y = f(\Theta) + \mathcal{B} \pm Noise \tag{1}$$

where $\mathcal{B}$ is the bias. The function $f(\Theta)$ can be obtained from SVR. A margin of tolerance ϵ is considered in SVR to accommodate the noise in the function [27].

For an independent variable $\theta_i \in \Re, \forall\, i = 1, \cdots, N$ and dependent variable $y \in \Re$, an ϵ-insensitive SVR problem can be formulated as a constrained optimization problem. An interested reader may refer [28] for the detailed formulation of SVR. The final primal form of the regression problem with l number of training data points is

$$\begin{aligned} \min \quad & \tfrac{1}{2}\mathbf{W}^T\mathbf{W} + C\sum_{j=1}^{q}(\zeta_j + \zeta_j^*) \\ \text{s.t.,} \quad & \zeta_j, \zeta_j^* \geq 0;\ \forall\, j = 1, \cdots, l \\ -\epsilon - \zeta_j^* \leq \ & y_j - \{\mathbf{W}^T\phi(\Theta) + b\} \leq \epsilon + \zeta_j;\ \forall\, j = 1, \cdots, l \end{aligned} \tag{2}$$

where $\mathbf{W}^T\phi(\Theta) + b$ represents the N-dimensional hyperplane, which is optimized within the limit of $[-\epsilon, \epsilon]$. $\phi(\Theta)$ is the Kernel function. The Kernel function transforms the low-dimensional input space into a high-dimensional feature space [28], hence converting a non-linear regression problem into a linear regression problem in the feature space. $\phi(\Theta) = \Theta$ represents a linear Kernel function. Such linear Kernel functions are also used in the proposed methodology in Section 3 due to

their simplicity. ζ_j and ζ_j^* in (2) are the slack variables, introduced to represent the fitting error, and the positive control parameter C defines the maximum penalty, which can be imposed on fitting errors. Various methods are available in the literature to efficiently obtain the control parameters of SVR [29] for regression analysis. The FDI model for PMU measurements are discussed next.

2.2. False data in PMU measurements

Erroneous instruments or cyber attacks can cause false or malicious data in PMU measurements. The model adopted in this paper for injecting false data in PMU measurements is described next.

2.2.1. Unintentional false data

The introduction of unintentional false data in PMU measurements is similar to the introduction of bad data in SCADA based measurements, mostly due to faulty current and potential transformers and other erroneous instruments (such as GPS). Also, it is stated in [30] that the noise in the PMU measurements ranges in $45-55$ dB. Thus, any random variation in the PMU measurements higher than $45-55$ dB can be accounted for unintentional false data in the PMU measurements.

2.2.2. Intentional false data

Contrary to the unintentional FDIs, intentional false data in PMU measurements can be injected by carefully crafted cyber attacks. These FDIAs are designed in a way so that the traditional bad data detection strategies [31] are bluffed. Hence, in this paper, intentional FDIs are considered as their detection is difficult as compared to the unintentional false data. Also, it is to be noted that various FDIA techniques exist in the literature. However, in this paper, a local FDIA [32] is considered, which focuses on injecting false data in a PMU by limiting the attacking region around it, as shown in Fig. 1. A brief description of the localized intentional FDIA approach is discussed next.

The model of the AC SE [7], which relates the measurements to the state variables, can be stated as $\mathbf{Z} = \mathbf{h}(\mathbf{X}) + \mathbf{e}$, where $\mathbf{Z}$ is the measurement vector containing all the field measurements, $\mathbf{X}$ is the set of state variables, function vector $\mathbf{h}(\cdot)$ relates $\mathbf{Z}$ and $\mathbf{X}$, and $\mathbf{e}$ represents the Gaussian error in the measurements $\mathbf{Z}$. The sensitivity matrix in SE that defines the dependency between $\mathbf{Z}$ and $\mathbf{X}$ is

$$\Delta\mathbf{Z} = \mathbf{H}\Delta\mathbf{X} \tag{3}$$

where $\mathbf{H}$ is the Jacobian matrix of AC SE. It is to be noted that the use of DC power flow equations of the power network may result in a linear DC SE. However, the use of AC power flow equations can result in a nonlinear AC SE or a linear AC SE, depending on the type of the measurements that are being used in the SE exercise. If only the SCADA based measurements are used in SE, then a nonlinear AC SE model needs to be solved for the system states [21]. However, if the measurements are being directly taken from the optimally placed PMUs [33], then even though the power flow equations are nonlinear, the overall SE exercise is based on a linear AC SE model [34]. It is because the PMUs directly give the bus voltage magnitudes and the associated phase angles. Thus with PMUs, one can directly get the system states, and the linear AC SE acts as a filter for any error or noise present in the PMU measurements. If the measurements from SCADA and PMUs are used in SE, then a nonlinear AC SE model needs to be solved for the system states [35].

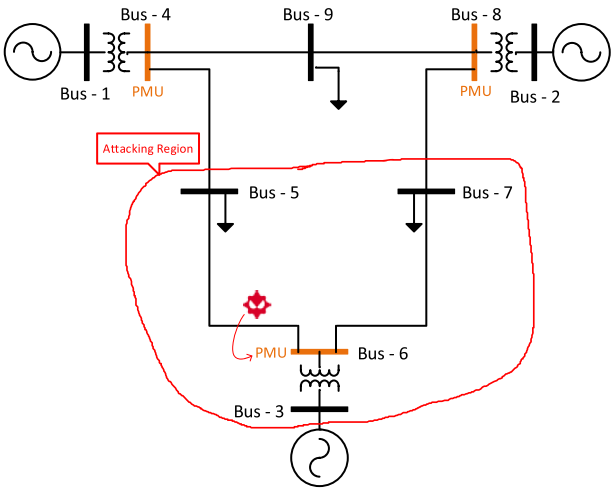


Fig. 1. Illustration of the attacking region for a PMU during intentional FDIA.

In order to have an intentional FDI in a PMU measurement (let it be the state variable X_1, which is the voltage magnitude or associated phase angle of bus 6 in Fig. 1 and is directly measured by the PMU at bus 6), an attacking region is created, as shown in Fig. 1. The attacking region includes the target PMU bus as well as all adjacent (directly connected) buses. Therefore, the measurement vector $\mathbf{Z}$ can be divided into two vectors; $\mathbf{Z}_1$, which includes all the measurements in the attacking region that directly affect the estimation of state X_1 in SE, and $\mathbf{Z}_2$, which includes the other measurements for SE of the rest of the system. Thus, (3) can be rewritten as

$$\begin{bmatrix}\Delta\mathbf{Z}_1\\ \Delta\mathbf{Z}_2\end{bmatrix} = \begin{bmatrix}\mathbf{H}_{11} & \mathbf{H}_{12}\\ \mathbf{H}_{21} & \mathbf{H}_{22}\end{bmatrix}\begin{bmatrix}\Delta X_1\\ \Delta\mathbf{X}_2\end{bmatrix} \tag{4}$$

where $\mathbf{H}_{11}$, $\mathbf{H}_{12}$, $\mathbf{H}_{21}$, and $\mathbf{H}_{22}$ are the sub-matrices of Jacobian $\mathbf{H}$ in (3). These depend on the state variable vector $\mathbf{X}$. In fact in (4), $\mathbf{H}_{21} = 0$ as per the definition of the attacking region, where the state X_1 is unaffected by change in measurement vector $\mathbf{Z}_2$. $\mathbf{X}_2$ in (4) are the state variables of the rest of the system. Injection of false data modifies $\mathbf{Z}_1$ to $\mathbf{Z}_1^{bad}$, which finally results in false estimation of state $\hat{X}_1$ in SE as $\hat{X}_1^{bad}$. The post-attack Largest Normalized Residue (LNR) from SE can be calculated as

$$LNR_{bad} = \|\mathbf{Z}^{bad} - \mathbf{h}(\hat{\mathbf{X}})\| = \left\| \begin{matrix}\mathbf{Z}_1^{bad} - \mathbf{h}(\hat{X}_1^{bad}, \hat{\mathbf{X}}_2)\\ \mathbf{Z}_2 - \mathbf{h}(\hat{\mathbf{X}}_2)\end{matrix} \right\| \tag{5}$$

where the state X_1 in (5) does not directly appear in function vector associated with $\mathbf{Z}_2$, as per the attacking region definition in (4). Further, the essential condition for FDIA to remain hidden from bad data detection phase in SE is

$$LNR_{bad} \approx LNR \Rightarrow \left\|\mathbf{Z}_1^{bad} - \mathbf{h}(\hat{X}_1^{bad}, \hat{\mathbf{X}}_2)\right\| \approx \|\mathbf{Z}_1 - \mathbf{h}(\hat{\mathbf{X}})\| \tag{6}$$

where $\hat{\mathbf{X}}$ is the true vector of the state variables without FDIA and LNR is the associated largest normalized residue from AC SE. The proposed approach, which is based on the use of SVR, for the detection and correction of false data in PMU measurements is discussed next.

3. Methodology

Generally, the deployment of PMUs in the electric power network aids in reducing the vulnerability of the power network against the FDIAs. However, recent literature shows that even PMUs are not completely secured against the novel cyber attacks [5,6,14,15]. Also, a meticulously designed FDIA tampers the PMU measurements (and the RTUs, if required) of the attacking region for a successful FDI in the PMU measurements, as discussed in Section 2.2.2. The conventional AC SE based false data detection algorithms cannot detect such type of FDIAs, as these FDIAs satisfy (6) and lead to an insignificant change in the measurement residue vector of AC SE. Also, the existing detection algorithms use the RTUs (SCADA based measurements) in the conjuncture of PMU based measurements to detect false data in the PMUs. However, different sampling rates of RTUs and PMUs makes the use of RTU measurements questionable for false or malicious data in PMU measurements.

Hence, a possible approach to detect such FDIAs is the one that does not use the SCADA based measurements for identifying false data in PMUs and also should not follow the conventional AC SE based measurement residual testing methodology. Furthermore, such an approach should not use the measurements from the attacking region, as they are logically tampered to satisfy Kirchoff's laws for successful FDIA. Hence, an SVR based machine learning approach is proposed in this paper for the real-time detection and correction of false data in single and multiple PMU measurements. The offline training of the SVRs aids in identifying an affine relationship between the voltage measurements from the sparsely placed PMUs in the power network. This developed affine relationship is then used for the real-time testing of the PMU measurements for detection and correction of intentional as well as unintentional false data in PMUs, even when the network is not fully observable by the PMUs alone. The proposed approach, being a machine learning-based approach, neither requires the information related to the parameters of the network topology nor the RTU/ SCADA based measurements. The AC model of the power network, based on which the affine relationships between the voltages of PMU measurements are developed, and the associated proposed approach for obtaining these relationships for real-time detection and correction of false data in PMU measurements are discussed next.

3.1. AC system model

For a n bus AC power network, the nodal equations can be written using the Ohm's law as

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1n} \\ Y_{21} & Y_{22} & \cdots & Y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n1} & Y_{n2} & \cdots & Y_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} \tag{7}$$

where I_i and V_i are the net current injection and voltage at the ith bus, respectively. $\mathbf{Y} \in \mathbb{C}^{n\times n}$ represents the bus admittance matrix of the AC power network with the $(i,j)^{\text{th}}$ element being Y_{ij}. If there are m PMUs installed in the system, then (7) can be rewritten as

$$[\mathbf{I}] = [\mathbf{Y}^{P} \quad \mathbf{Y}^{NP}] \begin{bmatrix} \mathbf{V}^{P} \\ \mathbf{V}^{NP} \end{bmatrix} \tag{8}$$

where $\mathbf{V}^{P} \in \mathbb{C}^{m\times 1}$ and $\mathbf{V}^{NP} \in \mathbb{C}^{(n-m)\times 1}$ are the voltage vectors for PMU and non-PMU buses, respectively. The admittances between the various PMU buses and non-PMU buses are represented by the associated bus admittance sub-matrices $\mathbf{Y}^{P} \in \mathbb{C}^{n\times m}$ and $\mathbf{Y}^{NP} \in \mathbb{C}^{n\times(n-m)}$. $\mathbf{I} \in \mathbb{C}^{n\times 1}$ is the net current injection vector. (8) can be further rewritten as

$$\mathbf{Y}^{P}\mathbf{V}^{P} + \mathbf{Y}^{NP}\mathbf{V}^{NP} - \mathbf{I} = 0 \tag{9}$$

If $(\mathbf{Y}^{NP}\mathbf{V}^{NP} - \mathbf{I})$ can be replaced by the associated mean vector $\mathbf{B}$ and variance vector $\boldsymbol{\delta}$, then (9) can be stated as

$$\mathbf{Y}^{P}\mathbf{V}^{P} + \mathbf{B} \pm \delta = 0 \tag{10}$$

where (10) represents an affine relationship between the voltages of the PMU buses only. It is to be noted that this affine relationship is still unknown, as $\mathbf{B}$ and $\boldsymbol{\delta}$ in (10) are yet to be known. The conventional deterministic approach to find $\mathbf{B}$ and $\boldsymbol{\delta}$ in (10) is to make use of the historical real-time/ dynamic SE [2] solutions from the respective mean and variance vectors of $\mathbf{Y}^{NP}\mathbf{V}^{NP}$ and $\mathbf{I}$ The issue with this conventional deterministic approach is that it requires RTU/ SCADA based measurements as well as PMU measurements. It also requires the information of the network parameters as well as the topology, which thus defeat the very purpose of an approach that does not require these information. Thus, the conventional deterministic approach to find $\mathbf{B}$ and $\boldsymbol{\delta}$ in (10) is not used in the proposed work due to its restrictive use.

Alternatively, the left hand side of (10) is analogously similar to the right hand side of (1), where Θ in (1) is analogous to the vector of the voltages of the PMU buses ($\mathbf{V}^{P}$). Also, by use of proper Kernel functions, (2) can be solved with known number of training data points to find (1) in SVR, as in Section 2.1. Thus, by offline training of SVR, the affine relationship between the voltages of the PMU buses in (10) can be statistically extracted by using only the PMU measurements (unlike the conventional deterministic approach). However, as per (10), this statistical extraction of affine relationship from SVR still requires the information of $\mathbf{Y}^{P}$ (the bus admittance sub-matrix between the PMU buses only). The means of obviating this requirement in the proposed approach (so that no network topology and parameter information is required) is discussed in the next section. Further, after obtaining the affine relationships through offline training of SVRs, the validation of these relationships in the real-time testing of SVRs can easily aid in detecting any possible false data in the PMU measurements.

3.2. Proposed approach

In the previous section, it is shown that an affine relationship between the voltages of the PMU buses can be obtained by SVR. The proposed approach first obtains this affine relationship by offline training of SVRs, using historically recorded PMU measurements. Then, the real-time PMU measurements are validated against this obtained affine relationship. Any deviation of the fresh PMU measurements, beyond an acceptable limit from the hyperplane of the SVR, possibly represents the presence of false data in the PMU measurements. The details of the proposed approach for detection of false data in a single PMU as well as multiple PMU measurements, and their subsequent correction are explained next.

3.2.1. False data in a single PMU

We first discuss our proposed approach in the case of false data in a single PMU measurement for the sake of clear understanding and basis of our approach. The case of false data in multiple PMU measurements is discussed in the next section.

A PMU provides the voltage phasor at a bus (voltage magnitude as well as the associated phase angle of the bus voltage), whereas SVR only works with real numbers. To deal with complex voltages, two SVR models are constructed, one for the voltage magnitudes and the other one for the associated voltage phase angles. Hence, for a power system with m number of PMUs, $2m$ number of SVR models are constructed to obtain the affine relationship between the m PMU voltage phasor measurements. In a power network with large number of PMUs, i.e., m being high, the strategy discussed in Section 3.3 is to be used.

Each SVR model is trained in a way so that it can linearly express the ith PMU bus voltage measurement in terms of other $(m-1)$ PMU bus voltage measurements. From (10), this relationship (superscript P is dropped for the sake of clarity) can be stated as

$$V_i = -\left[\sum_{\substack{j=1 \\ j\neq i}}^{m} Y_{ij}V_j + B_i \pm \delta_i\right] Y_{ii}^{-1} = \sum_{\substack{j=1 \\ j\neq i}}^{m} (w_{ij}V_j) + b_i \pm \sigma_i \;\forall\; i = 1, \cdots, m \tag{11}$$

where $w_{ij} = -Y_{ij}/Y_{ii}$, $b_i = -B_i/Y_{ii}$ and $\sigma_i = -\delta_i/Y_{ii}$. The relationship in (11) simply represents the hyperplane of the SVR whose parameters w_{ij}, b_i, and σ_i are to be determined by the successive offline training of each SVR model using the historical PMU measurements. This offline training of each SVR aids in the statistical equivalence of $\mathbf{B}$ and $\boldsymbol{\delta}$ (the mean and variance vector of $\mathbf{Y}^{NP}\mathbf{V}^{NP} - \mathbf{I}$ in (9)) and $\mathbf{Y}^{P}$ in (10). By doing so, the requirement of the information of the network topology and parameters and the RTU/ SCADA based measurements in the proposed approach is completely obviated. Post offline training, the SVR models are used to predict the bus voltage phasors ($\mathbf{V}^{Predict}$) of m PMU buses using the measured voltage phasors ($\mathbf{V}^{Meas}$) in real-time, which in turn allows the detection of intentional as well as unintentional false data in the PMU measurements. The obtained linear relationship in (11) aids in this purpose.

It is observed through various simulations that if the voltage measurement of pth PMU bus is modified by η%, then the impact of this erroneous measurement on the prediction of ith PMU bus voltage using (11), causes a prediction error which is always less than η%. Hence, if the predicted bus voltage ($\mathbf{V}^{Predict}$) is compared with the associated measured counterpart (i.e., $\mathbf{V}^{Meas}$), then the maximum deviation between the predicted and the measured bus voltages is observed for the pth bus. This observation acts as the core concept of the proposed work. The maximum deviation, thus obtained, is validated against a threshold for the determination of false data in the PMU measurements. This threshold value depends on the training parameter (tolerance ϵ in (2)) of the SVR model, and it aids in deciding the minimum quantum of false data that can be detected by the proposed algorithm.

The proposed approach is divided into two steps, the *detector* step and the *corrector* step. The *detector* step determines the presence of false data in the PMU measurements, by validating measurements against the trained SVR models, while the *corrector* step identifies the PMU(s), which produces corrupted measurements, as well as the amount of data corruption in the associated measurements from the identified PMU(s).

In the *detector* step, the measured bus voltage (magnitude as well as the associated phase angle) of each PMU bus is compared with its corresponding predicted voltage (obtained from the SVR model). If the deviation between the predicted and the measured voltages is more than a particular tolerance (outside the range of $\pm$ 0.5% in the proposed approach), then it is a sign of data corruption in the PMU measurements, and the PMU bus showing maximum voltage deviation is identified as the PMU generating suspicious measurements. The presence of suspicious measurements further activates the *corrector* step of the proposed approach.

Assuming that the ith bus is identified as the suspicious PMU bus in the *detector* step, then the *corrector* step involves verifying the suspicion and if found true, then subsequent correction of the false data in the suspicious PMU measurement. For verification, the measurement of the ith PMU in $\mathbf{V}^{Meas}$ vector is corrected by $V_i^{Predict}$, and the new measurement vector after such correction (denoted as $\mathbf{V}^{Corr}$, $\mathbf{V}^{Meas} \longrightarrow \mathbf{V}^{Corr} | V_i^{Meas} = V_i^{Predict}$) is again validated against the SVR models of (11). The output of the SVR models (denoted as $\mathbf{V}^{Corr_Predict}$) is compared with $\mathbf{V}^{Corr}$, and if **Corr_Error** (where, $\mathbf{Corr_Error} = \mathbf{V}^{Corr_Predict} - \mathbf{V}^{Corr}$) is within the acceptable range ($\pm$ 0.5% for this work), then this represents that the measurement of the ith PMU bus contains the false data, as it was suspected in the *detector* step, and the amount of false data is equal to ($V_i^{Meas} - V_i^{Predict}$). However, if **Corr_Error** is beyond the acceptable range, then this indicates FDI in measurements of multiple PMU buses.

The flowchart of the proposed approach for the real-time detection and correction of false data in PMU measurements is shown in Fig. 2. The use of the proposed approach to detect false data in multiple PMUs and to identify the corrupted PMUs along with the magnitude of false data is discussed next.

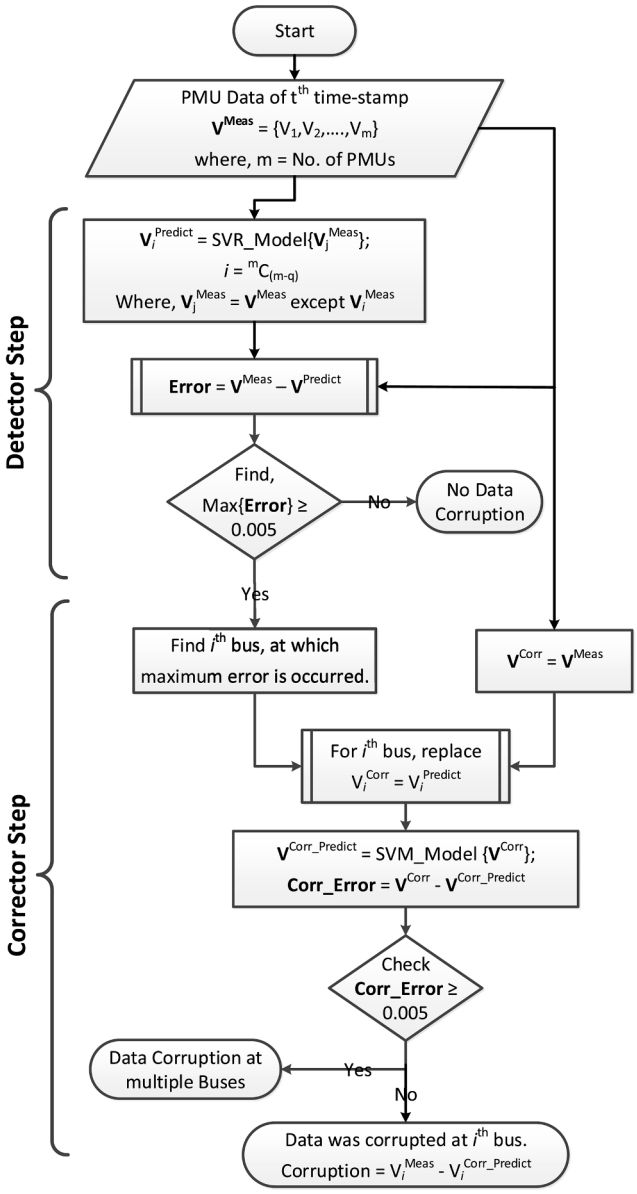


Fig. 2. Flowchart of the proposed approach.

3.2.2. False data in multiple PMUs

The proposed approach can also be easily used to detect false data in multiple PMU measurements and to correct these identified suspicious measurements. Based on the experience of operating the system, the system operator decides a threshold value q, where $q < m$ is the maximum number of PMUs that can be corrupted at any given point of time. More discussion on the value of q is given in Section 3.3. In such a scenario, the SVR model of (11) uses $(m - q - 1)$ PMU bus measurements, and ${}^{m}C_{(m-q)}$ SVR models are trained parallelly in an offline mode for the real-time (testing) detection and correction of false data in the multiple PMU measurements. The rest of the approach remains the same as in the case of false data in a single PMU, which is summarized in Fig. 2, i.e. the ${}^{m}C_{(m-q)}$ SVR models provide the predicted voltage phasors $\mathbf{V}^{Predict}$ of all PMU buses, which are then validated against the associated measured voltage phasors $\mathbf{V}^{Meas}$ to detect any possible data corruption in PMU measurements. Simulation results in Section 4.3 prove the effectiveness of the proposed approach in the case of corruption/ presence of false data in multiple PMU measurements.

3.3. Applicability of the proposed approach

Being a statistical and offline training based machine learning approach, it can be easily observed that the proposed approach is not computationally burdensome (during the testing phase) and the computation time is invariant of the system size. After offline training of the proposed SVR models, the proposed approach is tested in real-time for the detection and correction of false data in PMU measurements. The approach is capable of detecting and accurately correcting false data in PMU measurements, irrespective of the instance of the occurrence of bad data or an intentional/ unintentional FDI. Also, it is recommended to use successive training of the SVR models to accommodate the quasi-static as well as the dynamic changes in the operating conditions of the system [36]. This is generally true in any machine learning-based application to be trained for the changing system conditions and reducing the chance of giving false outputs [36]. The PMU measurements, which

are flagged as uncorrupted during the *detector* step, can be used for this successive retraining of SVR models so that these models represent the current system operating conditions. However, there can be a scenario where a significant change in the network topology or total load/generation occurs in the system in a very small period of time. In such a scenario, successive training alone may not be able to update the SVR models, which may lead to false alarms. Hence, to deal with such a situation, the system operator uses the proposed approach in conjunction with the topology processor and solution from AC SE [7], where any significant change in the system operating condition down weighs the response of the proposed approach to allow the retraining of SVRs. This also triggers the (successive) retraining of the SVR models for the new operating condition [36].

Also, in a power network with a large number of PMUs, i.e., m being a large number, the number of SVR models that need to be trained in the proposed approach, may be very high. In order to reduce the number of SVR models in the proposed approach, the power network is divided into multiple local areas, and then the SVR models are built by using the PMU measurements of the respective local area [37,38]. If required, distributed computing and parallel processing can also be used on lines similar to their use in AC SE [39]. It is to be noted that the multi-area and distributed computing application of the proposed approach reduces the number of SVR models that need to be trained in the offline mode, and this does not affect the real-time implementation of the proposed methodology. Further, based on the system operation experience, a system operator can choose the threshold q (the maximum number of multiple PMUs that can be corrupted in Section 3.2.2). The works in [40,41] can also be used to choose q (based on the fact that in the event of an intentional FDIA, the attack effort, in terms of multiple PMUs to be corrupted, has to be as minimum as possible for the FDIA to remain undetected).

4. Simulation results

The effectiveness of the proposed approach is tested on two test systems, the IEEE 118 bus system and the 246 bus NRPG of India. For the proposed approach, the control parameters of (2) are selected as $\epsilon = 0.005$ and $C = 1$, so as to simulate a soft margin SVR model with a maximum tolerable of $\pm$ 0.5%. A linear Kernel function $\phi(\Theta) = \Theta$ in (2) is used here for all the SVR models, due to simplicity. However, if required, other Kernel functions can also be used [28].

4.1. Test systems

Brief descriptions of the test systems used in this paper as well as their training and testing database creation are discussed next.

4.1.1. IEEE 118 bus system

The IEEE 118 bus system represents a part of the American Electric Power System, as of December 1962, and consists of 19 generators [42]. Optimal location of PMUs for the IEEE 118 bus system are obtained using linear integer programming [43] and are also validated with the results of [33]. The training and testing database for the IEEE 118 bus system are created by doing Optimal Power Flow (OPF) through MATPOWER [44] under different loading conditions in every 15s on a 24hr load profile (shown in Fig. 3).

In a practical scenario, the measurements from the PMUs generally contain noise. Hence, white Gaussian noise is added to the voltage phasor measurements of PMU buses to obtain similar measurement signals as obtained from practical PMUs. The study in [30] shows that the Signal to Noise Ratio (SNR) of $45 - 55$ dB is a good approximation to make a synthetic signal analogous to the measurement signal obtained from the PMUs. Hence, in this approach, SNR is kept at 55 dB for the bus voltage magnitudes and 50 dB for the associated voltage phase angles while maintaining the Total Vector Error (TVE) to be less than 1%. Out of the obtained 5760 time-samples generated using OPF, 30% are used for the purpose of real-time testing while the remaining are used for offline tranining of the SVRs.

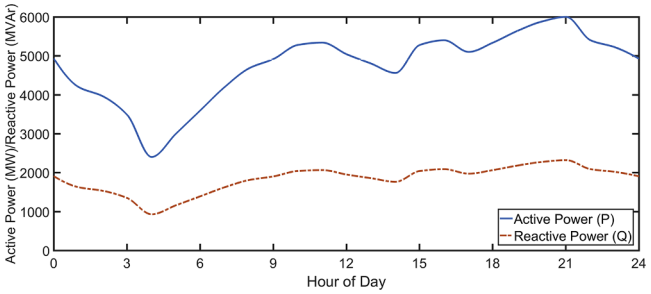


Fig. 3. Load profile for the IEEE 118 bus system.



Fig. 4. Location of PMUs in NRPG of India as on 21st April 2014.

4.1.2. Northern regional power grid (NRPG) of India

NRPG is the largest regional grid of India, which geographically covers seven states and two union territories of northern India [45]. The proposed approach is also tested on the actual measurements of the NRPG data. PMU measurements of 13 PMU locations, as shown in Fig. 4, of 21st April 2014 from 08:34:14 AM to 09:34:16 AM are made available for this study.

Out of the 13 PMU buses, the measurements of 3 buses (Bassi, Vindhyachal, and Moga) are partially unavailable. Hence, measurements of only 10 PMU buses (Kanpur, Hissar, Agra, Karcham, kishenpur, Dadri, Meerut, Balia, Bawana, and Rihand) are considered in this work.

4.2. Results for false data in single PMU

4.2.1. Results for the IEEE 118 bus system

Out of 118 buses, 32 buses are selected for complete system observability using [43]. Hence, 64 SVR models are trained, one each for each bus voltage magnitude and one each for each bus voltage phase angle using the training database, and then the effectiveness of the proposed approach is verified with the testing database. Data corruption on different PMU buses are simulated using the FDI model discussed in Section 2.2 for any randomly selected PMU bus in a way so that at any given time, only one PMU measurement is corrupted.

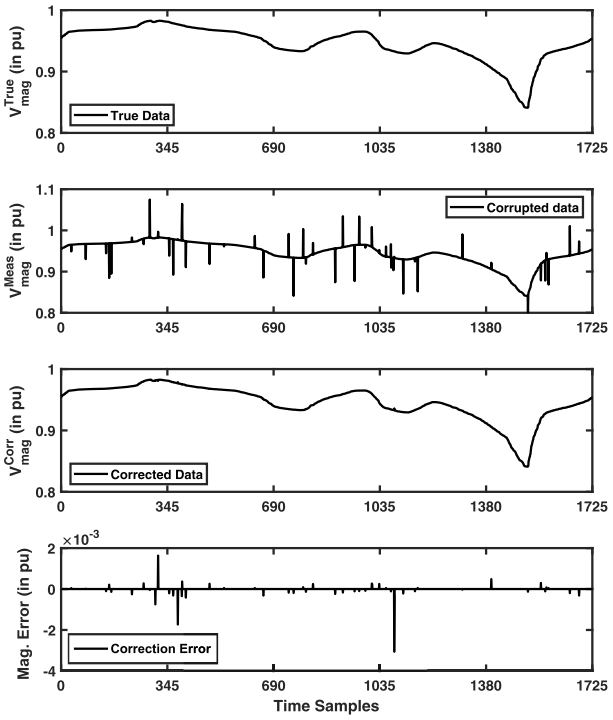


Fig. 5. Performance of the proposed approach for voltage magnitude at bus 75 in IEEE 118 bus system.

Magnitude Correction To assess the effectiveness of the proposed approach for detection and correction of false data in voltage magnitude measurements, the regression function of SVR for PMU buses is formulated as $\mathbf{W}^T|\mathbf{V}|^{PMU} + Bias \leq \epsilon$. A magnitude error in the range of $\pm$ 0.001% is injected in the different PMU measurements and then subsequently corrected with the proposed approach. Out of 32 PMUs for this test system, the maximum correction error is observed on the PMU bus 75. Hence, the detailed performance of the proposed approach for bus 75 is shown in Fig. 5, where uncorrupted voltage magnitude measurement is shown as *true data*, corrupted measurement is shown as *corrupted data*, output of corrector step is represented by *corrected data*, and the error between *true data* and output of the corrector step is shown as *correction error*. As shown in Fig. 5, the *correction error* is always less then $\pm$ 0.005*pu*, which justifies the accuracy of the proposed approach.

The performance of the proposed approach is also verified from Fig. 6, which shows the false data detection status (FDI detected as 1 and FDI not detected as 0) with respect to the range of error in between $\pm$ 0.001%. It can be seen from Fig. 6 that the proposed approach detects the magnitude error accurately in the range of $\pm$ 0.001%, except for the narrow margin of $\pm$ 0.000035%. This narrow range is termed as the *Alarm Insensitive Error Range*, which depends upon the chosen tolerance (ϵ) used in the offline training of the SVR models. This range can be further minimized by choosing a further

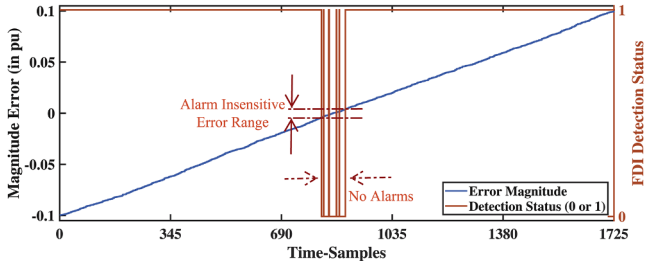


Fig. 6. Magnitude error vs false data detection status in IEEE 118 bus system.

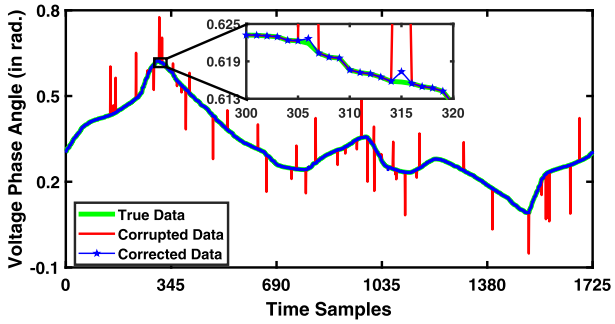


Fig. 7. Performance of algorithm for voltage phase angle at bus 75 in the IEEE 118 bus system.

small ϵ, as per the operator's experience based on system operation and the susceptibility of the PMU measurements to FDI [40,41].

Phase Angle Correction To assess the effectiveness of the proposed algorithm for detection and correction of false data in voltage phase angles, the regression function of SVR in terms of phase angle is chosen as $\mathbf{W}^T\delta^{PMU} + Bias \leq \epsilon$. Random angular error within the range of $\pm$ 0.1754*rad* is injected in different PMU buses. Out of the 32 PMU buses, the maximum correction error is observed on PMU bus 75. Hence, the detailed performance of the proposed approach for bus 75 is shown in Fig. 7, where the corrected data is compared with the corrupted data and true data. The deviation between the uncorrupted phase angle and output of the corrector step is shown in Fig. 8, which is always within the range of $\pm$ 0.01*rad*, and thus validates the performance of the proposed approach.

4.2.2. Results for the NRPG of India

Simulation results presented in the previous section justify the proposed approach for synthetic test systems. The performance of proposed approach on a real-world system is also tested. SVR models, for voltage magnitude as well as phase angle prediction, are trained (in an offline manner) with the 4000 time-samples for each of the PMU buses. The errors are created on different PMU buses in a way so that at any given time, only one PMU measurement is corrupted in this test system.

Magnitude Correction A voltage magnitude error in the range of $\pm$ 0.001% is simulated on different PMU buses at different time instants, and these erroneous measurements are then processed through the proposed approach. Voltage magnitude error between the true data (i.e., the voltage magnitude measurements before adding bad data) and the corrected data (i.e., the voltage magnitudes after processing through proposed approach) is given in Fig. 9, which presents the performance of proposed approach for correction of false data in voltage magnitudes of different PMU buses in the NRPG of India. Each of the boxplot in Fig. 9 provides visualization of 5 statistical properties. The top and bottom of the boxplot are the 25th and 75th percentiles, while the notch inside the box represents the median of the error range. The maximum and minimum values of the *correction error* for each PMU

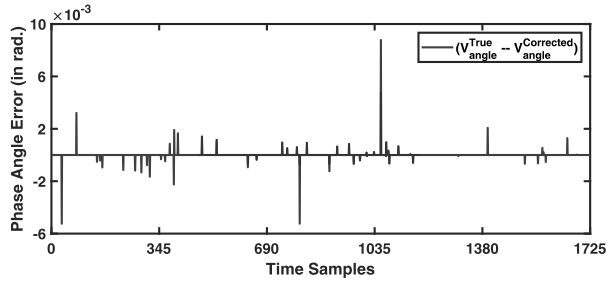


Fig. 8. Correction error between true data and corrected data at bus 75 in the IEEE 118 bus system.

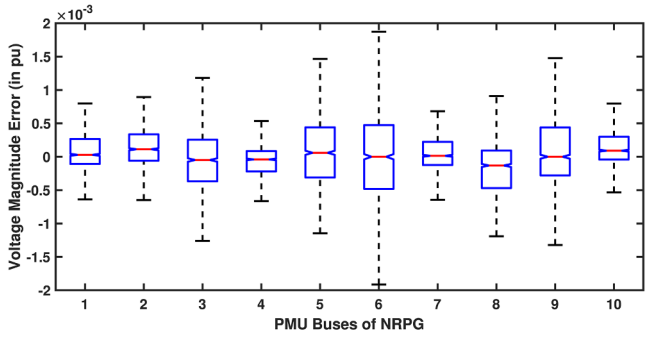


Fig. 9. Comparative performance of proposed approach for detection of false data in voltage magnitude of PMU buses in NRPG of India.

bus are represented by the whiskers above and below the box, respectively.

It can be observed from Fig. 9 that the correction error between the true measurements and the output of the proposed method is always less the $\pm$ 0.002*pu*, with few outliers, for voltage magnitude measurements of each PMU bus. One can also observe that the maximum correction error has occurred at the 6th PMU, which is located at *Dadri bus*. Hence, the individual performance of the proposed approach for *Dadri bus* is given in Fig. 10, which shows the measured voltage magnitude (i.e., *true data*), *corrupted data* after injecting false data in *true data*, and *corrected data* after processing *corrupted data* through the proposed approach.

The deviation between the true voltage magnitude data and associated corrected data for *Dadri bus* is shown in Fig. 11, which proves the effectiveness of the proposed approach to detect and correct the false data in voltage magnitude measurements of a real-world grid.

Phase Angle Correction A voltage phase angle error in the range of $\pm$ 0.1754*rad* is simulated on different PMU buses of the NRPG at different time samples. The comparative performance of the proposed approach for the detection of false data in phase angle on various PMU buses of NRPG system is shown in Fig. 12.

It can be observed from Fig. 12 that the worst performance is obtained in 6th PMU, i.e., *Dadri bus*. Hence, the individual performance of the proposed approach for *Dadri bus* is shown in Fig. 13, where *corrected data* represents the phase angles after correction of false data, whereas the *true Data* represents the initial phase angle measurements without false data. *Correction Error* between *true data* and *corrected data* for phase angle at *Dadri bus* is shown in Fig. 14.

4.2.3. Result analysis

Numerical simulations have been performed on IEEE 118 bus system and NRPG of India for the detection of false data in voltage magnitude and associated phase angles of PMU measurements and their subsequent correction. Metrics indicating the performance of the proposed approach, for the worst-performing bus in synthetic as well as

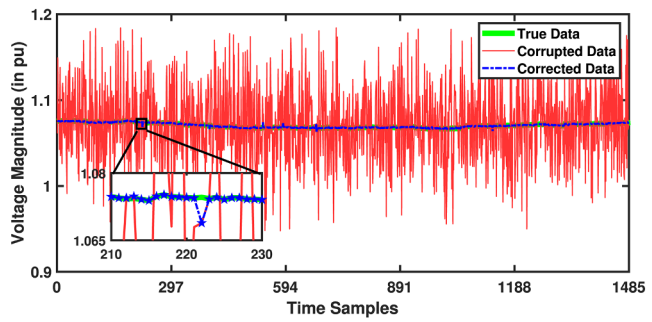


Fig. 10. Performance of proposed approach for voltage magnitude at *Dadri Bus* in NRPG of India.

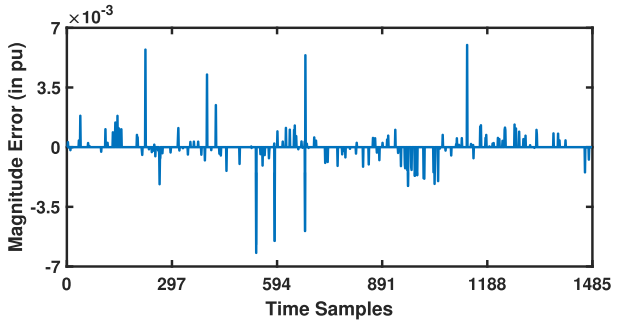


Fig. 11. Deviation between the true voltage magnitude and corrected output at *Dadri Bus* in NRPG of India.

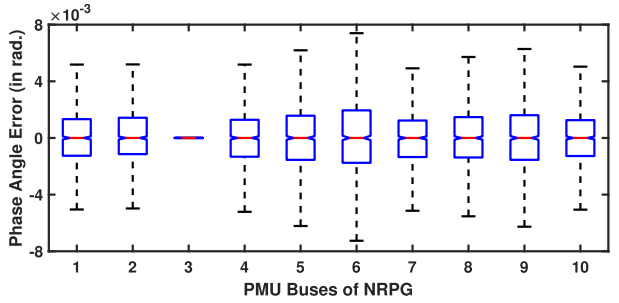


Fig. 12. Comparative performance of proposed approach for detection of false data in phase angle of PMU buses in NRPG of India.

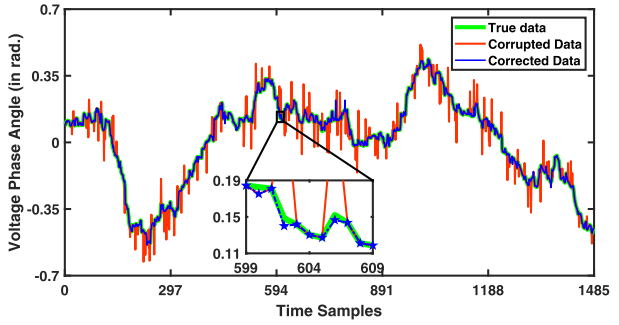


Fig. 13. Performance of proposed approach for voltage phase angle at *Dadri Bus* in NRPG of India.

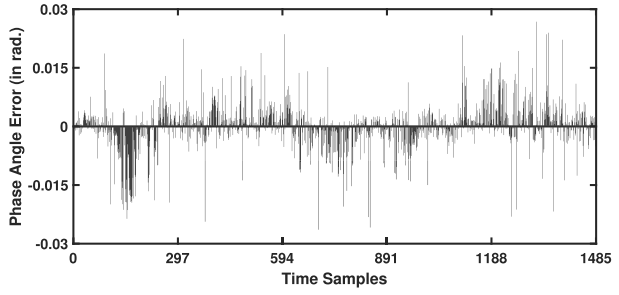


Fig. 14. Deviation between the true voltage phase angle and corrected data at *Dadri Bus* in NRPG of India.

real-world power systems are summarized in Table 1. The deviation between the *corrected data*, which is the output of proposed approach and the *true data* are presented in terms of *RMS error, mean error*, and *standard deviation* in the correction error.

It can be observed from Table 1 that the RMS of the effective *correction error* for voltage magnitude correction is always within the range of $\pm$ 0.001%, and the same for voltage phase angle is always in the

Table 1
Performance comparison of proposed approach for false data detection in a single PMU.

Metrics	IEEE 118 bus system		NRPG of India	
	Volt. Mag. Error(*pu*)	Volt. Ang. Error(*rad*)	Volt. Mag. Error(*pu*)	Volt. Ang. Error(*rad*)
Mean	-7.7735×10^{-6}	2.7994×10^{-5}	-2.6105×10^{-5}	-1.2873×10^{-4}
RMS	3.1146×10^{-4}	1.0450×10^{-3}	8.1329×10^{-4}	5.5229×10^{-3}
SD	3.1145×10^{-4}	1.0449×10^{-3}	8.1315×10^{-4}	5.5233×10^{-3}

range of $\pm$ 0.006*rad* (i.e., $\pm$ 0.35°), and hence always maintaining TVE below $\pm$ 1%. Also, these values are expected to remain invariant of the system size. More details on the real-time performance of the proposed methodology is discussed in Section 4.4.

4.3. Results for false data in multiple PMUs

Section 4.2 shows the detailed performance of the proposed approach, on synthetic as well as the real-world system, for the case where only one PMU measurement is corrupted. However, the proposed approach can also be used for the case, where multiple PMUs are sending false measurements to the control center. Hence, to verify the effectiveness of the proposed approach for multiple PMU corruption scenario, the proposed approach is also tested on the NRPG of India for the case, where 2 PMUs out of 10 are sending false measurements to the control center.

A random magnitude error in the range of $\pm$ 0.001% and the phase angle error in the range of $\pm$ 0.1754*rad* is simulated on the NRPG data-set in a way so that for any given time-sample, only two PMUs have corrupted measurements. Fig. 15 shows the distribution of *correction error* of voltage magnitude measurements for different PMUs after processing through the proposed approach.

Similarly, Fig. 16 shows the distribution of *correction error* of voltage phase angles for various PMUs buses, after the corrupted voltage phase angle measurements of the PMU buses are processed through the proposed approach.

It can be observed from Figs. 15 and 16, that maximum *correction error* has occurred on *Dadri bus*, while correcting the multiple false data in voltage magnitude and phase angle measurements. Hence, Table 2 summarises the performance of the proposed approach on *Dadri bus* using various indexes, and validating the proposed approach for multiple PMU corruption scenario.

4.4. Comparison with related work in [8,9,11,15,16,23]

Most of the existing false data detection strategies do not consider the intelligence of the attacker to modify the RTU/ SCADA based measurements, which is consistent with the physical Kirchoff's voltage and current laws. Hence, the motivation behind this paper was to propose a approach, which can detect the false data in PMU measurements without relying on the use of SCADA based measurements. To the best of the author's knowledge, no other paper discusses the use of sparsely placed PMUs for false data detection in PMUs and their subsequent correction, except [11]. However, [16] discussed the approach for detection of FDI in PMUs using a set of secured meters, and [15] discussed the use of ADMM for recovery of the PMU measurements using the temporal spatial correlation among multiple PMU measurements. Hence, numerical simulations has been carried out on the IEEE 118 bus test system to compare the detector and corrector step of proposed approach with [15,16], and the results are presented in Table 3.

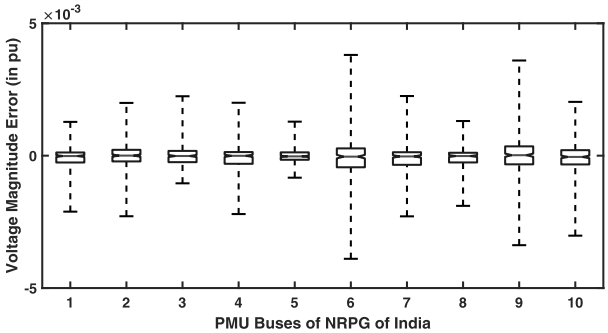


Fig. 15. Performance of algorithm for the case where magnitude measurements of 2 PMUs are corrupted in NRPG of India.

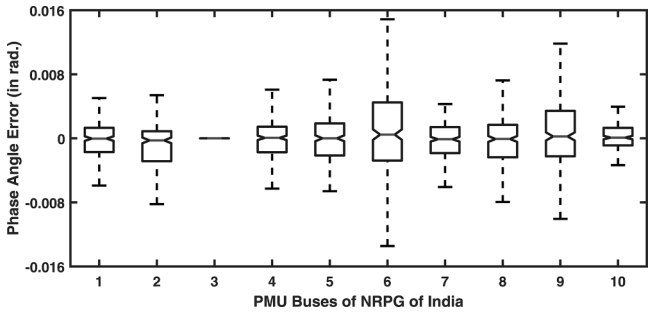


Fig. 16. Performance of algorithm for the case where phase angle measurements of 2 PMUs are corrupted in NRPG of India.

Table 2
Performance of proposed approach for *Dadri bus* of NRPG system for multiple PMU corruption scenario.

Metrics	Voltage Magnitude Error (in *pu*)	Voltage Phase Angle Error (in *rad.*)
Mean	-3.5250×10^{-5}	3.4188×10^{-4}
SD	0.0016	0.0072
RMS	0.0016	0.0072

Table 3 compares the performance of the proposed approach with [15,16] in terms of false alarms during the *detector step*, and the mean errors during the *corrector step*. It can be observed, that number of false alarms in the proposed approach are nominal as compared to the number of false alarms in [16]. This is because FDIA can occur on the set of secured meters of [16], whereas the proposed approach raises false alarms when prediction error is equal to the tolerance value of SVR models (*i.e.*, $V^{Meas} - V^{Predict} = \epsilon$). However, the false alarms in our proposed approach get detected and corrected during the *correction phase*, because of the *correction error* becoming less then $\pm$ ϵ. Furthermore, the mean values of *correction error* for proposed approach and the method in [15] are also presented, and it can be observed that error is minimal in the proposed approach as compared to [15]. This is because the selection of penalty factor (ρ) in ADMM in [15] is a challenge. Also, it has been explained earlier in Section 1 that [16] can only be used to detect the presence of false data, whereas ADMM [15] can be modified to identify the magnitude of false data in the measurement set. Hence, the correction phase and the detection phase are not implemented for methods in [15,16], respectively in Table 3.

A comparison between the proposed method and the claims made in

Table 3
Comparison of detector and corrector step of proposed approach with other methods for IEEE 118 bus test system.

	% of false alarm during detector step		Mean error during corrector step	
	$\lvert V_{mag}\rvert$	$\angle V_{ang}$	$\lvert V_{mag}\rvert$ (in pu)	$\angle V_{ang}$ (in rad.)
Proposed Approch	0.126%	0.253%	-2.6105×10^{-5}	-1.2873×10^{-4}
Projection Statistics [16]	82.36%	85.57%	-	-
ADMM [15]	-	-	7.135×10^{-3}	8.324×10^{-2}

the related approaches of [8,9,11,23] is also summarized.

- *Application* - The proposed approach, which is based on the use of SVR, is computationally efficient and can easily be used for the real-time application (after successful offline training), similar to the approach as in [23]. The approaches in [8,9] are computationally expensive and are not fit for real-time applications, whereas the approach in [11] is mainly proposed for offline applications.
- *Performance* - A SVR model trained with ample amount of training database (as is the case in the proposed approach ensured by the measures discussed in Section 3.3) is able to detect and correct false data at all times. In contrast, the approach in [8] fail to detect false data in case of a low probability of occurrence. We implemented the work presented in [23], and it turns out that the work in [23] fails to detect false data 30% of the time. Also, the works presented in [9,23] are windowing based techniques that are susceptible to raise false alarms.
- *Adaptability* - In case, the system conditions are incredibly different from the instance at which SVRs were trained (such as a change in system topology or load or generation change), the proposed approach requires successive retraining to adapt to such conditions (on lines of discussion in Section 3.3), which is also true for the approaches proposed in [9,23].
- *Accuracy* - As shown in the results, the worst-case RMS and mean errors for the voltage magnitudes and voltage phase angles correction in the proposed approach are always less than 0.0027*pu* and 0.0064*rad*, respectively, which are less than the errors reported in [8,9,11].

5. Conclusion

An SVR based approach for real-time detection and correction of FDIs in the PMU measurements is proposed in this paper. Most of the existing approaches in the literature do not consider the intelligence of attackers to formulate an FDIA consistent with the physical network laws. Hence, they fail to detect stealth attacks using the RTU/ SCADA based measurements. The proposed approach, however, is useful in this case, as verified by the simulation results on the synthetic data of the IEEE 118 bus system and the real-world data of NRPG of India. An affine relationship between the voltage phasors of the PMUs is statistically obtained by offline training of SVRs through the use of historical PMU measurements and AC model of the power network. No SCADA measurements or network topology information is required in the proposed approach. The affine relationships developed by the trained SVRs are then tested in real-time to detect and correct false data in PMUs. Proper care is also given to handle the changing system conditions in the training database of the SVRs by use of successive retraining, and the AC SE solutions. The overall performance of the proposed approach is found to be extremely satisfactory, through numerical simulations carried on synthetic as well as real-world data as well as through comparisons with few recent approaches in the literature. The smallest false data in the measurements that can be detected using the proposed algorithm depends on the free parameters (ϵ and C) of the SVR models. In present work, these parameters are selected as $\epsilon = 0.005$ and $C = 1$ to formulate a soft margin SVR model, which accurately detects the false data in single as well as multiple PMU measurements. Effective error when correcting the false data in voltage magnitude and phase angle measurements are always observed in the range of $\pm$ 0.002*pu* and $\pm$ 0.008*rad.*, respectively, while maintaining the TVE less than 1%.

CRediT authorship contribution statement

Gaurav Khare: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing - original draft. **Abheejeet Mohapatra:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing - review & editing. **S.N. Singh:** Funding acquisition, Project administration, Resources, Software, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.epsr.2020.106866.

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